WORKSHEET 9.4

Free-Response Type: Rates and Accumulation

Unit 9: Exam Strategy and Free-Response Types

Key ideas

- An integral of a rate is an amount. Its units are the units of the rate times the units of time.
- Amount at time t = starting amount + integral of (rate in − rate out).
- The derivative of a rate tells how fast the rate itself is changing: units per time, per time.
- For a greatest or least amount, compare the endpoints with every time at which rate in = rate out.

Worked examples

Example 1 · An amount

A bath fills at $R(t) = 20 - 2t$ litres per minute for $0 \le t \le 10$. How much water enters in the first 5 minutes?

Solution. $\displaystyle\int_0^5 (20 - 2t)\,dt = 100 - 25.$

75 litres.

Example 2 · An average rate

Find the average rate at which the bath fills for $0 \le t \le 10$.

Solution. $\displaystyle\frac{1}{10}\int_0^{10} (20 - 2t)\,dt = \frac{100}{10}.$

10 litres per minute.

Example 3 · The derivative of a rate

Find $R'(4)$ and interpret it.

Solution. $R'(t) = -2$.

The rate of filling is decreasing at 2 litres per minute per minute.

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Practice

A calculator is required on this page. Give decimal answers to three places.

CORE

1. $r(t)$ is measured in people per hour and t in hours. What are the units of $\displaystyle\int_2^5 r(t)\,dt$?

2. With the same units, what are the units of $r'(t)$?

3. A tank holds 200 litres at $t = 0$. Water enters at $f(t)$ and leaves at $g(t)$ litres per minute. Write an expression for the amount in the tank at time t.

4. Explain in words the meaning of $\dfrac{1}{5}\displaystyle\int_0^5 r(t)\,dt$, where $r(t)$ is in people per hour.

EXAM LEVEL

5. Visitors enter a gallery at $E(t) = 60te^{-0.3t}$ people per hour and leave at $L(t) = 25 + 3t$ people per hour, for $0 \le t \le 9$. There are 100 visitors at $t = 0$. How many visitors enter for $0 \le t \le 9$?

6. Is the number of visitors in the gallery increasing or decreasing at $t = 6$? Give a reason.

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7. Find $E'(2)$, and interpret it in context with units.

8. Find the greatest number of visitors in the gallery for $0 \le t \le 9$. Justify your answer.

Question 9 · Free-response practice

Parcels arrive at a depot at the rate $A(t) = 150 + 60\sin\left(\frac{\pi t}{6}\right)$ parcels per hour and are sorted at a constant rate of 160 parcels per hour, for $0 \le t \le 8$. At $t = 0$ there are 200 unsorted parcels.

(a) How many parcels arrive for $0 \le t \le 8$?

(b) Is the number of unsorted parcels increasing or decreasing at $t = 4$? Give a reason.

(c) Find the greatest number of unsorted parcels for $0 \le t \le 8$. Justify your answer.

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Worked solutions

1. People.
2. People per hour per hour.
3. $200 + \int_0^t \big(f(x) - g(x)\big)\,dx$.
4. The average rate, in people per hour, over the first 5 hours.
5. $\int_0^9 E(t)\,dt \approx 500.893$, about 501 visitors.
6. Increasing: $E(6) \approx 59.508 > L(6) = 43$.
7. $E'(2) \approx 13.171$. At $t = 2$ hours, the rate at which visitors enter is increasing at about 13.171 people per hour per hour.
8. $E(t) = L(t)$ at $t \approx 0.517$ and $t \approx 7.494$. Numbers: 100 at $t = 0$; about 93.910 at $t \approx 0.517$; about 266.427 at $t \approx 7.494$; about 254.393 at $t = 9$. The greatest is about 266 visitors.
9. (a) $\int_0^8 A(t)\,dt \approx 1371.887$ parcels.
   (b) Increasing: $A(4) - 160 \approx 41.962 > 0$.
   (c) $A(t) = 160$ at $t \approx 0.320$ and $t \approx 5.680$. Unsorted parcels: 200 at $t = 0$; about 198.405 at $t \approx 0.320$; $200 + \int_0^{5.68} \big(A(t) - 160\big)\,dt \approx 370.778$ at $t \approx 5.680$; about 291.887 at $t = 8$. The greatest is about 371 parcels.

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